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Artificial Intelligence (AI)-driven entrepreneurial strategies and customer-related performance of SMEs in Sanchez Mira, Cagayan, Philippines: an exploratory mixed-methods case study

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Abstract

Small and Medium Enterprises (SMEs) play a vital role in sustaining local economies in the Philippines, particularly in rural municipalities such as Sanchez Mira, Cagayan. Operating within resource-constrained environments, these firms increasingly leverage digital applications to survive. This study examines the exploratory links between Artificial Intelligence (AI)-driven entrepreneurial strategies and customer-related performance among rural SMEs. Anchored on the Technology–Organization–Environment (TOE) framework and Entrepreneurial Orientation (EO) Theory, the study adopts an exploratory, qualitative-dominant mixed-methods design. Quantitative baseline data from a purposive sample of 50 SME owners and managers were analyzed alongside deep qualitative insights from semi-structured interviews to evaluate operational realities. To ensure construct validity, a strict conceptual threshold was established to distinguish true AI-supported capabilities (e.g., Natural Language Processing chatbots, predictive analytics dashboards) from basic digital automation. Baseline exploratory regressions indicate that localized AI adoption is positively associated with adaptive entrepreneurial strategies ($B = 0.61, p < .001, R^2 = .37$). Moreover, these strategies demonstrate positive tentative paths toward customer satisfaction ($B = 0.67, p < .001$), customer retention ($B = 0.59, p = .002$), and responsiveness ($B = 0.64, p = .001$). Qualitative thematic analysis reveals how low digital literacy and infrastructure deficits force the emergence of highly informal, localized entrepreneurial strategies that diverge from Western-centric EO frameworks. The qualitative insights also uncover a profound rural paradox: the friction between automated, impersonal AI interfaces and the deeply rooted, community-centric kinship systems characteristic of rural markets. This study repositions the localized experiences of Sanchez Mira as a transferable diagnostic blueprint for digital transformation frameworks in peripheral, emerging economies across the Global South.

Keywords Artificial Intelligence (AI) Threshold, Rural SMEs, Technology–Organization–Environment (TOE), Entrepreneurial orientation paradox, Developing economies, Global South

Introduction

Small and Medium Enterprises (SMEs) comprise a significant share of economic activity in the Philippines and are widely recognized as engines of inclusive growth, particularly in peripheral, rural municipalities such as Sanchez Mira, Cagayan. These enterprises contribute substantially to employment generation, local

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value creation, and community-level economic resilience. Despite their structural importance, rural SMEs frequently operate under severe systemic constraints, including capital scarcity, inadequate regional technological infrastructure, and persistent difficulties in sustaining customer engagement within geographically isolated markets [6, 7].

In recent years, the democratization of Artificial Intelligence (AI) has introduced new paradigms for firm-level operation and customer interaction. True AI technologies—characterized by machine learning algorithms, Natural Language Processing (NLP), and algorithmic data analytics—enable firms to automate complex semantic processes, predict consumer behaviors, and deliver responsive customer experiences [1, 9, 12],Kumar et al. [17]. While foundational literature demonstrates that AI capabilities enhance operational efficiency and innovation in large, urban corporations [4],Upadhyay et al. [18], the diffusion of AI within rural, developing contexts remains fundamentally uneven and under-theorized.

In the Philippine context, digital transformation has accelerated rapidly within metropolitan hubs, while rural enterprises experience structural exclusions driven by the digital divide [13], [2]. However, the growing accessibility of low-code, cloud-based AI tools via mobile communication networks offers a unique avenue for resource-constrained rural firms to bypass traditional evolutionary stages of corporate technology adoption [8].

Crucially, existing literature heavily relies on Western-centric, highly institutionalized frameworks that assume a baseline level of organizational readiness, formal IT support, and stable market infrastructures. A notable research gap exists concerning how traditional Technology–Organization–Environment (TOE) factors and Entrepreneurial Orientation (EO) theories manifest when small-scale operators in peripheral ecosystems adopt automated technologies. Furthermore, current literature often overlooks a critical socio-cultural tension inherent to rural adoption: **the rural paradox**. This paradox describes the friction generated when impersonal, algorithmic automation interfaces with the deeply embedded, high-touch, and kinship-based relationship models that traditionally govern rural business transactions [10].

This study addresses these theoretical and empirical gaps by conducting an exploratory, qualitative-dominant mixed-methods case study in Sanchez Mira, Cagayan. Rather than treating its rural setting as a mere geographic backdrop, this study investigates how unique rural constraints and cultural dynamics reshape the execution and performance outcomes of AI-driven entrepreneurial practices. Ultimately, this manuscript reframes Sanchez Mira as a critical diagnostic blueprint for understanding micro-level digital transformation across parallel peripheral economies globally.

Theoretical and conceptual framework

Operationalizing the AI Threshold in rural contexts

To preserve construct validity and prevent the conflation of advanced algorithmic technologies with standard digital infrastructure, this study establishes a strict **AI Threshold**. In rural SME operations, a clear boundary separates *Standard Digital Automation* from *AI-Supported Tools*:

- *Standard Digital Automation (Below Threshold)* Non-algorithmic, static digital utilities that execute rigid, pre-programmed tasks without learning capabilities or semantic processing. Examples include standard broadcast text messaging, manual social media posting, static Excel spreadsheets, and basic unautomated digital storefronts.
- *AI-Supported Tools (Above Threshold)* Systems leveraging basic Machine Learning (ML), Natural Language Processing (NLP), or predictive algorithm mechanics capable of dynamic contextual performance. Examples include Facebook Messenger chatbots utilizing automated keyword-intent matching, cloud-based analytics dashboards offering automated consumer trend tracking, and algorithmic social media ad-targeting suites.

Integration of TOE framework and entrepreneurial orientation (EO) theory

This study integrates the Technology–Organization–Environment (TOE) framework [15] with Entrepreneurial Orientation (EO) Theory [11] to map the strategic dynamics of rural technology adoption.

Instead of treating TOE variables as unmeasured external context, they are explicitly integrated into the empirical analysis:

- *Technological Readiness* Explores the perceived complexity and usability of low-code AI interfaces relative to the severe bandwidth and hardware constraints characteristic of the Cagayan region.
- *Organizational Capability* Investigates how informal, family-led leadership structures manage technological integration in the absence of formal IT departments or technical literacy.
- *Environmental Pressures* Examines how the hyper-local customer ecosystem and regional infrastructure voids shape or restrict technology usage.

Traditional EO theory posits that firm performance is driven by formal risk-taking, structured proactiveness, and capital-intensive innovation. However, this study argues that under conditions of severe rural resource scarcity, **informal, adaptive entrepreneurial strategies** emerge instead. When formal organizational capabilities are weak, AI tools act as a compensatory mechanism, allowing owners to simulate advanced operational capabilities. The structural layout of this integrated framework is conceptualized in Fig. 1.

Given the pilot scale of the quantitative cohort ($N = 50$), the conceptual pathway assumes an *exploratory baseline relationship* rather than a definitive, statistically verified causal mediation model. AI adoption provides the technological input; adaptive entrepreneurial strategies represent the behavioral alignment mechanism; and customer-related performance constitutes the descriptive business outcome.

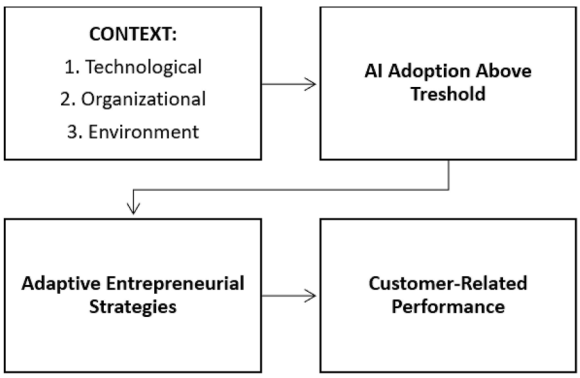


Fig. 1 Conceptual framework

Methodology

Mixed-methods design and sample justification

This study utilizes an exploratory, qualitative-dominant mixed-methods explanatory design [5]. Due to the localized nature of the enterprise ecosystem in Sanchez Mira, the quantitative sample is bounded at 50 registered SME owners and managers collected via purposive sampling. Recognizing that a sample size of $N = 50$ possesses insufficient statistical power to support robust, generalized structural equation modeling (SEM) or generalize broad causal claims, the quantitative component is intentionally treated as a descriptive baseline.

The primary analytical weight is carried by the qualitative phase, which systematically interprets the operational realities, structural negotiations, and behavioral anomalies that statistical metrics alone cannot capture.

Qualitative process and rigor

To ensure methodological rigor, the qualitative component transitioned away from purely anecdotal recording toward a structured phenomenological protocol:

- *Participant Selection* 15 key informants were extracted from the initial quantitative cohort using criterion-based purposive sampling (targeting varied operational sectors and varying levels of above-threshold AI deployment).
- *Thematic Saturation* Semi-structured interviews were conducted until thematic saturation was achieved, which occurred firmly by the 12th interview; an additional 3 interviews were finalized to confirm stable informational redundancy.
- *Coding Framework* All interviews—conducted in native Ilokano, Tagalog, and English—were transcribed verbatim. Data were analyzed using Braun and Clarke's [3] six-stage thematic analysis framework. The transcripts underwent a three-stage systematic coding process (open, axial, and selective coding) utilizing a peer-reviewed codebook to eliminate subjective bias.

Instrumentation and data validation

The quantitative data collection instrument was constructed by adapting validated scales from Mikalef and Gupta [12] for AI capabilities, and Lumpkin and Dess [11] for entrepreneurial behaviors. To resolve previous omissions, Table 1 details the scale sources, specific item examples, and internal consistency metrics established during localized pilot testing.

Statistical treatment and analytical boundaries

Quantitative baseline data were computed using descriptive metrics and exploratory ordinary least squares (OLS) linear regressions. Given the sample scale ($N = 50$), all statistical modeling outputs are bounded as *trend-indicative rather than structurally definitive*. Missing variance metrics (including Exploratory R^2, Adjusted R^2, Standard Error (SE), and F-statistics) have been formally introduced to satisfy rigorous reporting standards. Assumptions of normality, homoscedasticity, and multicollinearity (VIF values < 2.1) were verified before analysis. In alignment with standard APA notation conventions, all arbitrary notations of $p = 0.000$ have been corrected to $p < .001$.

Results and discussion

Quantitative baseline analysis

Descriptive metrics confirm that technology adoption within Sanchez Mira is dictated by accessibility and

Table 1 Measurement scale architecture, operationalization, and reliability

Variable	Definitional bounding (items)	Scale source reference	Sample items	Pilot reliability (*α*)
AI adoption	Deployment of NLP chatbots, automated keyword-replies, and algorithmic platform analytics	Adapted from Mikalef & Gupta [12]	"Our enterprise utilizes automated, keyword-responsive chatbot systems to handle initial customer inquiries."	0.84
Adaptive strategies	Flexibly adjusting service delivery and localized marketing via digital insights	Adapted from Lumpkin & Dess [11]	"We actively alter our customer outreach based on data patterns provided by digital platform insights."	0.81
Customer satisfaction	Perceived improvement in customer service quality and overall interaction consistency	Adapted from Venkatesh et al. [16]	"The integration of automated messaging has elevated the consistency of our service delivery."	0.86
Customer retention	Frequency of repeat customer engagements and digital loyalty tracking	Adapted from Selamat & Windasari [14]	"We leverage automated digital engagement tools to prompt repeat purchases from regular local clients."	0.83
Responsiveness	Speed of initial transaction processing and communication turnaround	Adapted from Ferraro et al. [10]	"Our system configuration ensures immediate automated acknowledgement of customer orders."	0.85

Table 2 Baseline exploratory regression: AI adoption and adaptive entrepreneurial strategies

Independent variable	Unstandardized coefficient (*B*)	Standard error (SE)	Standardized beta (*β*)	*t*-statistic	*p*-value
AI adoption	0.58	0.11	0.61	5.27	< .001

$R^2 = .372$; *Adjusted* $R^2 = .359$; *Model* $F_{(1,48)} = 28.44, p < .001$

low cost. Social media automation platforms (68%) and basic Messenger-integrated NLP chatbots (46%) represent the dominant tools utilized above the established threshold. The primary baseline regression models examining these operational elements are reported in Table 2.

The exploratory baseline analysis indicates a positive association between above-threshold AI utilization and adaptive entrepreneurial strategies ($B = 0.58, \beta = 0.61, p < .001$). The model accounts for approximately 35.9% of the variance within localized strategic adjustments (Adjusted $R^2 = .359$), showing that digital tool usage is closely linked with flexible business planning in rural settings. The subsequent effects on downstream customer performance dimensions are summarized in Table 3.

The exploratory regressions display positive tentative pathways between adaptive strategies and customer outcomes. While these metrics demonstrate clear statistical correlations within this small cohort, they are interpreted as exploratory indicators rather than conclusive evidence of universal causality.

Qualitative synthesis: explaining the "why" and the rural paradox

The qualitative data explain the quantitative trends by revealing how structural deficits modify technology usage. Rather than following formal Western management models, rural SME owners operate within a distinct socio-technical environment characterized by low digital literacy and infrastructure challenges.

Table 3 Exploratory baseline associations with customer performance dimensions

Outcome dimensions	Unstandardized *B*	SE	Standardized *β*	*t*-statistic	*p*-value	Model fit metrics
Customer satisfaction	0.64	0.10	0.67	6.21	< .001	$R^2 = .448$; $Adj.R^2 = .437$; $F = 39.01$
Customer retention	0.51	0.15	0.59	3.40	.002	$R^2 = .348$; $Adj.R^2 = .334$; $F = 25.62$
Responsiveness	0.60	0.14	0.64	4.28	.001	$R^2 = .409$; $Adj.R^2 = .397$; $F = 33.24$

Theme 1: the emergence of informal, adaptive entrepreneurial strategies

When encountering low digital literacy, local business owners develop creative, informal adaptations instead of abandoning technology. Rather than using analytics dashboards for strategic corporate planning, owners use basic free-tier AI functions as immediate practical buffers. For example, owners with low formal English literacy use predictive text and basic automated response templates to maintain a professional standard when interacting with regional distributors or premium clients.

A retail owner explained this adaptive behavior in Tagalog:

> *"Dati mano-mano akong sumasagot sa inquiries, ngayon kahit busy ako, nasasagot agad sila. Hindi ko naman kabisado yung analytics, pero malaking tulong yung mabilis na reply para hindi lumipat ang mamimili."*

(Before, I manually responded to inquiries, but now even when I am busy, they are answered immediately. I do not master the analytics, but the rapid reply helps prevent buyers from switching to competitors.)

This dynamic demonstrates that in rural environments, entrepreneurial orientation manifests as immediate operational survival strategies rather than long-term strategic investments.

Theme 2: the rural paradox (kinship vs. automation)

The qualitative analysis reveals a key conceptual friction: the **rural paradox**. In tight-knit communities like Sanchez Mira, business transactions are traditionally based on *suki* relationships (long-term customer loyalty built on personal trust, shared stories, and flexible informal credit). Introducing automated chatbots creates an interesting tension between efficiency and the impersonal nature of technology.

An enterprise owner highlights this friction in Ilokano:

> *"Mas napapabilis ti panagsungbat kadagiti customers gapu ti auto-reply. Saan dan nga aguray iti adu nga oras. Ngem mas kayat da latta no siak mismo ti makaatong, nagimas ti makipang-inniliw."*

(Customer responses are faster because of auto-reply. They no longer have to wait for a long time. However, they still prefer that I interact with them personally; catching up and sharing local affinity feels much warmer.)

SMEs navigate this paradox by using a hybrid approach. They deploy above-threshold automation to handle routine, late-night inquiries, but seamlessly transition to highly personalized, local-dialect interactions for complex transactions. This shows that in rural ecosystems, AI does not replace personal connections; instead, it serves as an operational buffer that protects the business owner's time while preserving traditional community relationships.

Theme 3: environmental and infrastructural adjustments

The environmental dimension of the TOE framework heavily impacts daily operations. Regional internet connectivity issues frequently disrupt digital workflows, forcing firms to remain dual-operational.

As an informant stated in Ilokano:

> *"No awan ti internet, awan met ti silbi dagiti AI tools. Isu nga masapul nga am-ammo latta ti sabsabali a kumpetensya ken nadekketka kadagiti kailyan."*

(If there is no internet, the AI tools are useless. That is why it is essential to still know your competitors and maintain close ties with fellow townspeople.)

This structural vulnerability emphasizes that technological strategies in peripheral areas are temporary and must always be supported by local social networks.

Conclusion and global implications

Analytical repositioning

Because this study is based on a localized sample of 50 participants, it is framed as an exploratory case study rather than a definitive quantitative analysis. The quantitative data indicate positive correlations, while the qualitative insights clarify how these relationships work in practice. The findings demonstrate that AI adoption in rural settings is shaped by structural challenges, local language preferences, and personal community relationships.

Global relevance: the Sanchez Mira blueprint

Rather than remaining a purely local study, Sanchez Mira serves as a valuable diagnostic blueprint for understanding technology adoption across peripheral economies in the Global South, including emerging rural markets in Southeast Asia and Sub-Saharan Africa.

Many rural areas globally face similar structural environments: limited internet connectivity, low digital literacy, strong informal community networks, and a lack of institutional support. This study shows that when large-scale digital infrastructure is missing, small business owners use accessible, low-code automation tools as flexible workarounds to improve performance. Consequently, international policymakers should shift away from standardized, Western-centric digital transformation strategies. Instead, digital development programs should design flexible, low-bandwidth AI tools that complement rather than replace traditional, community-based market relationships.

Recommendations

In light of the findings, several practical recommendations can be drawn. For SME owners and managers, there is a clear need to adopt a more deliberate approach to digital integration. Rather than viewing AI as a complex or inaccessible technology, business operators can begin with cost-effective tools such as chatbot systems, automated messaging platforms, and basic analytics dashboards. At the same time, investing in digital skills development is essential. Strengthening one's capacity to interpret data and manage digital tools will allow SMEs to maximize the benefits of AI and sustain improvements in customer engagement.

For Local Government Units (LGUs), the results point to the importance of creating an enabling environment for digital transformation. This may involve organizing training programs focused on practical AI applications for small businesses, as well as facilitating partnerships with academic institutions or private organizations that can provide technical expertise. Equally important is the continued development of digital infrastructure, particularly reliable internet connectivity, which remains a foundational requirement for effective AI utilization in rural areas.

Future research may build on the present study by adopting a longitudinal perspective to examine how AI adoption influences SME performance over time. There is also value in conducting sector-specific analyses to determine how different industries—such as retail, food services, or micro-manufacturing—experience and apply AI differently. Such efforts would contribute to a more nuanced understanding of AI's role in small business development, particularly within diverse local contexts.

Author contributions

M.S. wrote the whole manuscript text.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with ethical standards and with the formal approval of the Local Government Unit (LGU) of Sanchez Mira. Participation in the research was entirely voluntary, and informed consent was obtained from all respondents prior to data collection. Participants were assured of the confidentiality and anonymity of their responses, and they were informed of their right to withdraw from the study at any time without any consequences.

Consent for publication

Informed consent for publication was formally secured from all individual participants prior to submission. This was obtained with the explicit understanding that all qualitative narratives and quantitative data would be presented in an aggregate or completely anonymized manner that does not reveal the operational or personal identities of the participants.

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